

EECS 841: Computer Vision

Lecture 6

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Outline (Chapter 3)

Lecture 6_ Point Transform _Part I_ Color Image

Image Transformation: Different ways to transform an image into another image

- Geometric transform
- Point transform

Outline

Different ways to transform an image into another image

- Geometric transform
- Point transform of Grayscale image
- Point transform of Color Image
 - Multi-spectral transformation
 - Multi image transformation
 - change detection

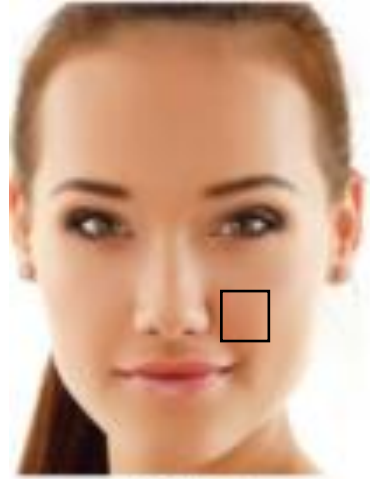
Multispectral Transformation

- Convert an RGB image to a grayscale image, which involves a mapping from a three-channel image to a single-channel image.
- The simplest approach would be to average the three values:

$$I'(x, y) = \frac{1}{3}(I_R(x, y) + I_G(x, y) + I_B(x, y))$$

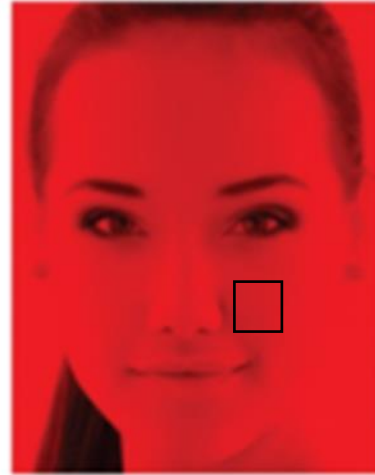
Multispectral Transformation

Figure 3.24 Top: Original RGB image and three separate color channels.



image

(R = ?, G = ?, B = ?)



red

(R = 255, G = 0, B = 0)



green

(R = 0, G = 255, B = 0)



blue

(R = 0, G = 255, B = 255)

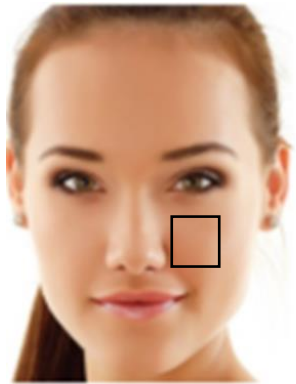
Can we see the brownish complexion if we choose same shade for R, G, B?

(R = 232, G = 190, B = 170)

RGB to Gray Scale Calculator	
232	190 170
Convert to Gray Scale Clear	
Equivalent Gray: (200, 200, 200)	Color & Gray: 

Multispectral Transformation

Multispectral Transformations - RGB Transformations (cont.)

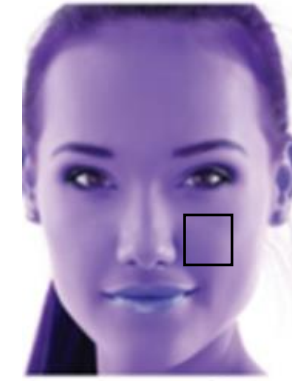


image

(R= 232, G = 190, B = 172)

RGB to Gray Scale Calculator	
232	190
172	
Convert to Gray Scale Clear	
Equivalent Gray: (200, 200, 200)	Color & Gray:  

Which two channels got
their intensity value
swapped to get this?



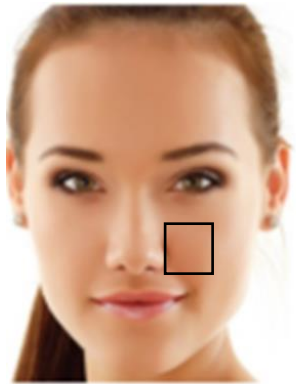
swapped1

(R= 172, G = 190, B = 232)

RGB to Gray Scale Calculator	
172	190
232	
Convert to Gray Scale Clear	
Equivalent Gray: (189, 189, 189)	Color & Gray:  

Multispectral Transformation

Multispectral Transformations - RGB Transformations (cont.)

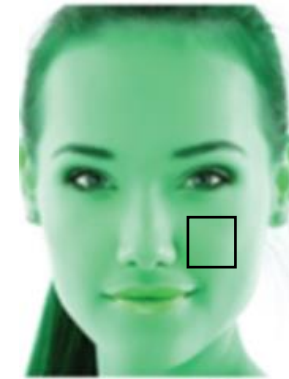


image

(R= 232, G = 190, B = 172)

RGB to Gray Scale Calculator		
232	190	172
Convert to Gray Scale		Clear
Equivalent Gray: (200, 200, 200)	Color & Gray:  	

Which two channels got
their intensity value
swapped to get this?



swapped2

(R= 190, G = 232, B = 172)

RGB to Gray Scale Calculator		
190	232	172
Convert to Gray Scale		Clear
Equivalent Gray: (213, 213, 213)	Color & Gray:  	

Multispectral Transformation

Pseudocolor

Multispectral Transformation

Pseudocolor

- **Pseudocolor:** process used to assign an RGB value to each gray level based upon some criterion other than simply replicating the gray level three times.

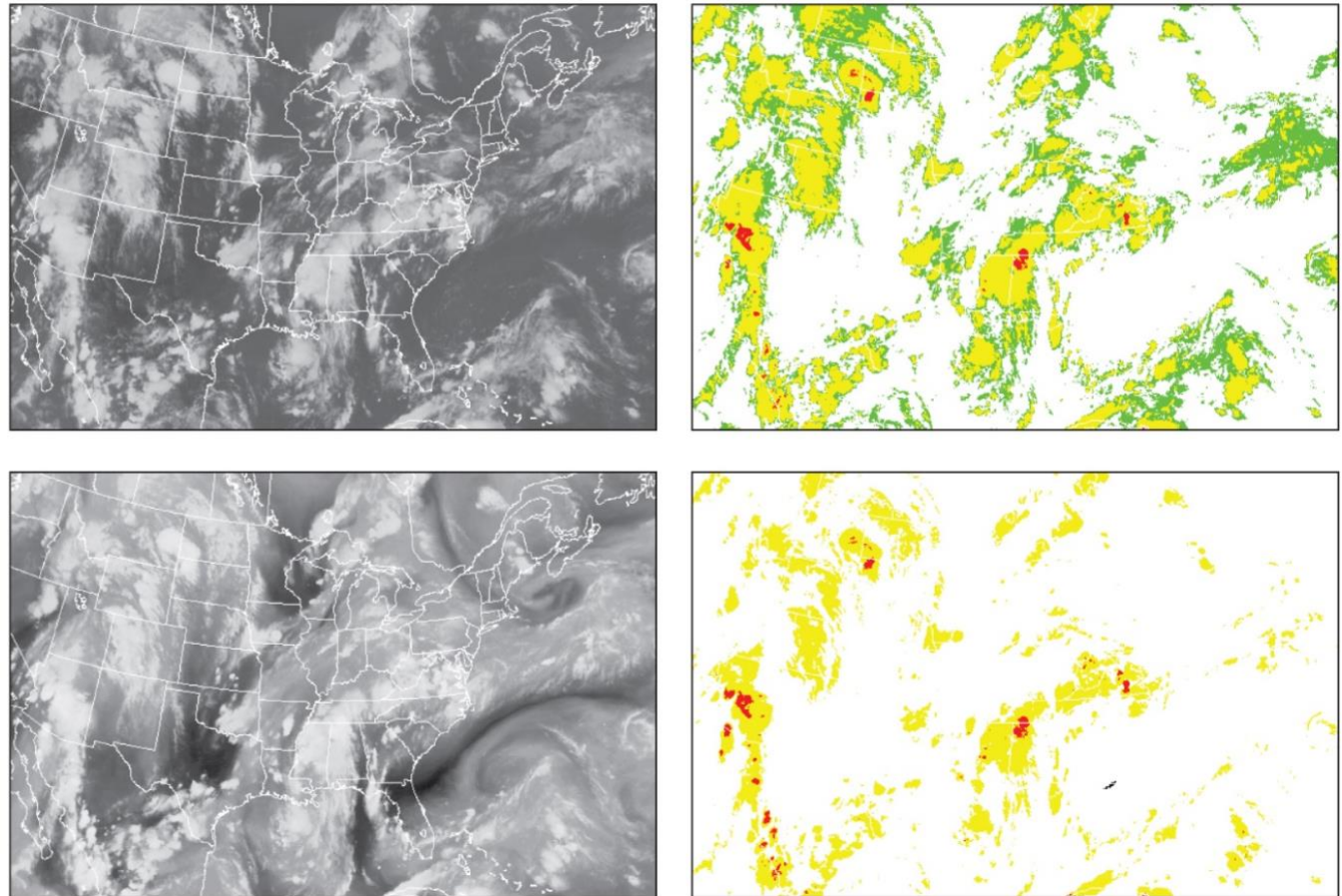
This process assigns colors not according to their actual appearance in the real world, so the resulting image is known as a false color image, or pseudocolor image.

Multispectral Transformation

Pseudocolor (cont.)

Figure 3.25

Pseudocolor display using density slicing of GOES infrared (top) and water vapor (bottom) images.



NOAA, <http://www.goes.noaa.gov>

Chromakey

- **Chromakey:** widely used in the movie and broadcasting industry to separate foreground from background for the purpose of blending multiple images.
- Historically blue was used, because blue is the farthest color from human skin color and because high-contrast film that was sensitive to only blue was widely available. The result, therefore, was known as **bluescreening**.
- The widespread use of digital video cameras that are more sensitive to green than to red or blue, combined with the use of digital video formats that also emphasize green over blue, have led to the dominance of using a green backdrop, known as **greenscreening**.

Chromakey (cont.)



Chromakey Fail



Multi-Image Transformations

Introduction to Video

Quick Notes about Video

- A video can be thought of a sequence of images (frames).
- The human eye can differentiate between individual images if images switch at a rate between 10-12 images per second (10-12 fps)
- Different Frame Rates Usage
 - 18 frames per second:* Early motion picture films.
 - 24 frames per second:* Worldwide standard for movie theater film projectors.
 - 300+ frames per second:* High-speed cameras for very slow-motion photography (often used for miniatures to make models seem larger on screen).
 - 2500+ frames per second:* Very high-speed cameras for special effects such as pyrotechnic photography and explosions.

Frames taken from a Video of Tennis Tournament



Multi-image Transformation

Multi-Image Transformations - Arithmetic Operations

- **Absolute difference:** computes for each pixel location the absolute value of the difference between the pixels:

$$I'(x, y) = |I_1(x, y) - I_2(x, y)|$$

- **Linear interpolation:** produces a convex combination of the two inputs:

$$I'(x, y) = \eta I_1(x, y) + (1 - \eta) I_2(x, y)$$

*(Used to produce a new image
Eta(n) is : 0 → 1 or 0 % → 100%)*

Change Detection

Change Detection - Frame Differencing

- **Difference image:** to compare successive image frames in the video sequence.

$$I'(x, y) = |I_t(x, y) - I_{t-1}(x, y)| > \tau$$

(in general)

Change Detection - Frame Differencing

- **Double-difference image:** used to solve the double-image problem, that is, the difference image will contain foreground pixels not only where the foreground object is located in the current frame but also where it was in the previous frame.

$$I' = |I_t - I_{t-1}| > \tau \quad \text{or} \quad |I_{t+1} - I_t| > \tau$$

- Visually, it shows the difference in terms of mapping both the pictures in the same window
- The difference will be captured from either t and t+1 moment or t and t-1 moment
- In the final output, it only shows one image which shows the difference of both the images

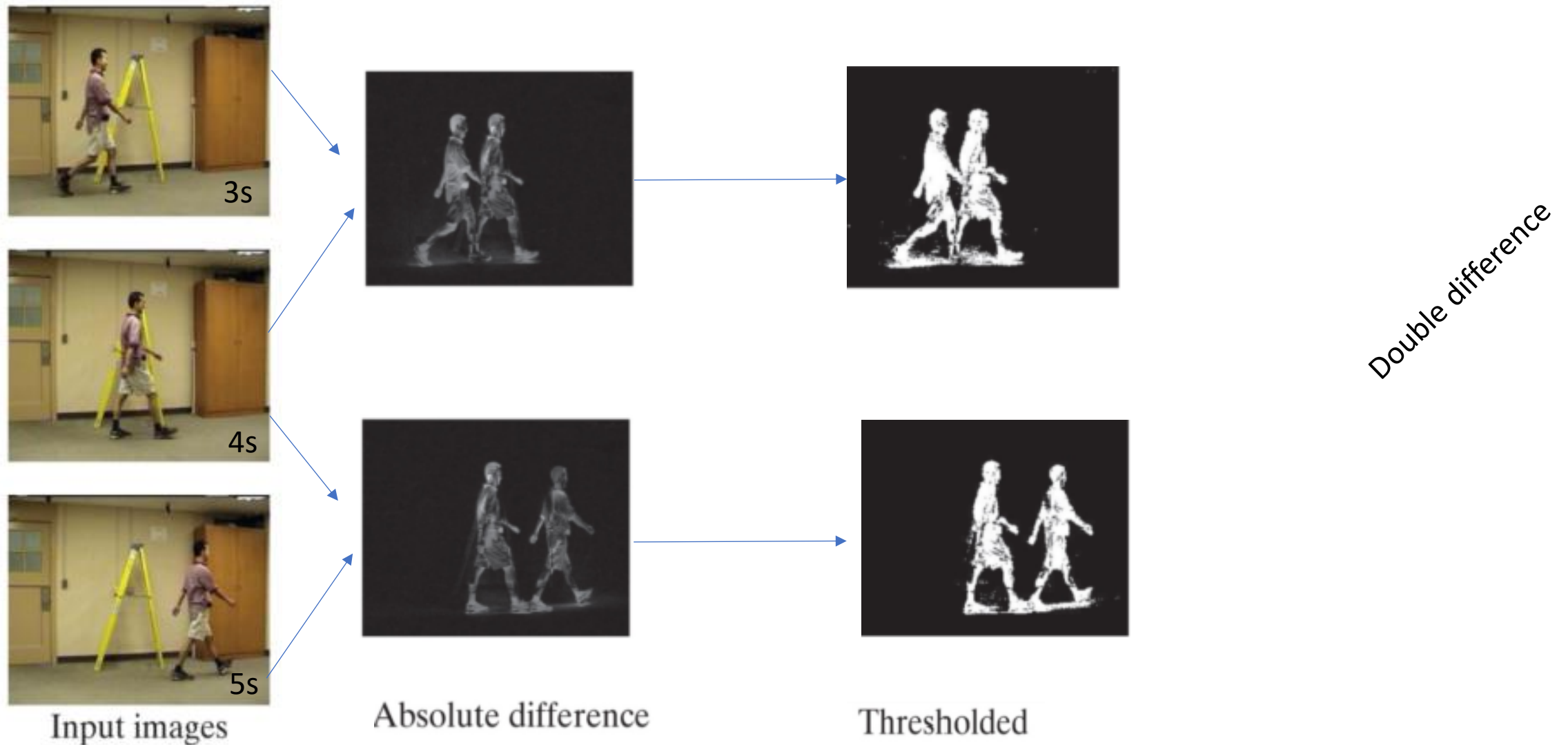
Change Detection - Frame Differencing

- **Triple-difference image:** combines the absolute differences from all three image pairs using addition and subtraction prior to thresholding:

$$I'(x, y) = (|I_{t-1} - I_t| + |I_{t+1} - I_t| - |I_{t-1} - I_{t+1}|) > \tau$$

- Visually, it shows the difference in terms of mapping both the pictures on $t-1 \rightarrow t+1$, in the same window
- The difference will be captured from 3 picture: $t-1 \rightarrow t \rightarrow t+1$
- In the final output, it only shows one image which shows the difference from 3 image

Visual Representation of Frame Differencing



Visual Representation of Frame Differencing



Input images



Absolute difference



Thresholded

Triple difference